

Human Retinal Organoids as a Platform for Evaluating Gene Delivery and Therapeutic Efficacy

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Gene-based therapies offer a promising strategy to restore cellular function in inherited retinal diseases (IRDs), particularly in patients who retain structural retinal integrity despite functional loss. However, clinical translation of gene therapy remains limited by the lack of delivery system capable of achieving efficient, cell type-specific targeting while minimising immune activation. Adeno-associated viral (AAV) vectors and lipid nanoparticles (LNPs) represent distinct gene delivery systems, with differing mechanisms of cargo delivery, expression profiles, and safety considerations; however, systematic and directly comparable evaluation in human-relevant models remains lacking.

To address this gap, we utilise human retinal organoids as a physiologically relevant platform for the systematic assessment of gene delivery strategies. We propose a stepwise pipeline: (i) optimisation and comparative screening of delivery systems in retinal organoids, incorporating clinically relevant administration routes (subretinal-like and intravitreal-like) to evaluate delivery efficiency, transgene expression, cell-type specificity, and vector distribution; (ii) application of optimised strategies in retinal disease models (e.g. X-linked juvenile retinoschisis (*RS1*) and retinitis pigmentosa (*LRAT*, *RPGR*)) to assess therapeutic efficacy and establish correlations between delivery performance and functional rescue; and (iii) incorporation of microglia into retinal organoids to model the retinal immune microenvironment and evaluate delivery-induced immune responses.

By integrating delivery optimisation, route modelling, disease-specific validation, and immune profiling within the human-based system, this platform provides a translational framework for the rational selection and development of gene therapies for retinal diseases.